

Resolvent, Scattering, and Heat Dynamics of a One-Dimensional Delta Interaction

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Abstract

We give a convention-complete solution of the one-center delta interaction on the real line in units $\hbar = 2m = 1$. Its pole produces exactly one attractive bound state; the remaining spectrum, left-right scattering, and odd/even channels are closed for every real coupling, including threshold and free faces.

1 The frozen singular Hamiltonian

For $\alpha \in \mathbb{R}$, consider the form

$$q_\alpha[\psi] = \int_{\mathbb{R}} |\psi'(x)|^2 dx + \alpha |\psi(0)|^2, \quad \mathcal{D}(q_\alpha) = H^1(\mathbb{R}). \quad (1)$$

The one-dimensional trace inequality makes point evaluation infinitesimally form bounded relative to the Dirichlet integral. Thus (1) is closed and lower bounded and defines one self-adjoint operator H_α . Integration by parts identifies its domain:

$$\mathcal{D}(H_\alpha) = \{\psi \in H^2(\mathbb{R} \setminus \{0\}) : \psi(0+) = \psi(0-), \quad \psi'(0+) - \psi'(0-) = \alpha\psi(0)\}, \quad (2)$$

and $H_\alpha\psi = -\psi''$ off zero. This normalization is the owner; changing the jump sign exchanges attractive and repulsive chambers.

Point interactions and their complete solvability are classical. The monograph of Albeverio, Gesztesy, Høegh-Krohn, and Holden contains a direct chapter on the one-center one-dimensional delta interaction [1]. Our contribution is the self-contained convention and executable closure below, not a priority claim.

2 One scalar equation gives the resolvent

For $\kappa > 0$, the free kernel at energy $-\kappa^2$ is

$$G_\kappa(x, y) = \frac{e^{-\kappa|x-y|}}{2\kappa}.$$

Theorem 1 (Rank-one resolvent). *If $2\kappa + \alpha \neq 0$, then*

$$(H_\alpha + \kappa^2)^{-1}(x, y) = \frac{1}{2\kappa} \left\{ e^{-\kappa|x-y|} - \frac{\alpha}{2\kappa + \alpha} e^{-\kappa(|x|+|y|)} \right\}. \quad (3)$$

Proof. Away from $x = y, 0$, both exponentials solve the homogeneous equation. The free term is continuous and its derivative has jump -1 at $x = y$. At zero, the kernel value in (3) is $e^{-\kappa|y|}/(2\kappa + \alpha)$. The image term has derivative jump $\alpha e^{-\kappa|y|}/(2\kappa + \alpha)$, exactly α times that value. It therefore satisfies (2) and the resolvent source equation. Uniqueness follows from invertibility away from the displayed pole. $\square$

The formula is a Krein rank-one correction. The exceptional equation $2\kappa + \alpha = 0$ has a positive solution precisely when $\alpha < 0$; it is spectral data, not a missing regular grid point.

3 Spectrum and two-channel scattering

The odd subspace automatically vanishes at zero, so H_α is free there. On the even subspace the interface becomes the half-line Robin condition $\psi'(0+) = (\alpha/2)\psi(0)$. The odd sine transform and the even Robin generalized eigenfunctions

$$\varphi_{\alpha,k}(x) = \sqrt{\frac{2}{\pi}} \frac{2k \cos(kx) + \alpha \sin(kx)}{\sqrt{4k^2 + \alpha^2}}, \quad x > 0, \quad k > 0,$$

give the two continuum transforms; when $\alpha < 0$, the even transform is completed by the single normalized bound vector in (4).

Theorem 2 (Complete sign atlas). *The essential spectrum is $[0, \infty)$ and is purely absolutely continuous; the singular-continuous spectrum is empty. If $\alpha < 0$, there is exactly one simple eigenvalue and normalized eigenfunction,*

$$E_b = -\frac{\alpha^2}{4}, \quad \phi_b(x) = \sqrt{-\frac{\alpha}{2}} e^{\alpha|x|/2}. \quad (4)$$

For $\alpha \geq 0$ there is no eigenvalue.

For momentum $k > 0$, a unit-amplitude wave incident from the left has

$$r_\alpha(k) = \frac{\alpha}{2ik - \alpha}, \quad t_\alpha(k) = \frac{2ik}{2ik - \alpha}. \quad (5)$$

The scattering matrix has odd eigenvalue 1 and even eigenvalue $(2ik + \alpha)/(2ik - \alpha)$, both of modulus one.

Proof. The resolvent pole gives (4); direct integration gives $\|\phi_b\|_2^2 = (-\alpha/2)(-2/\alpha) = 1$. Stone's formula applied to the explicit odd Dirichlet and even Robin resolvents yields the Lebesgue-density transforms displayed above. Their denominator $4k^2 + \alpha^2$ has no positive-real zero, the zero-energy solution is not in L^2 , and the only off-continuum singularity is the displayed attractive pole. Hence there is no embedded point or singular-continuous spectrum. For scattering, write $e^{ikx} + re^{-ikx}$ on $x < 0$ and te^{ikx} on $x > 0$. Continuity gives $t = 1 + r$ and the jump gives $2ikr = \alpha t$, proving (5). Finally

$$|r|^2 = \frac{\alpha^2}{4k^2 + \alpha^2}, \quad |t|^2 = \frac{4k^2}{4k^2 + \alpha^2},$$

so flux is one; diagonalizing the symmetric left-right matrix gives the two channel phases. $\square$

At $\alpha = 0$ the operator is exactly free. For nonzero coupling, the zero-energy limit is perfectly reflecting and the high-energy limit is transparent. The attractive pole occurs at $k = i|\alpha|/2$; a repulsive coupling has no upper-half-plane pole.

References

- [1] S. Albeverio, F. Gesztesy, R. Høegh-Krohn, and H. Holden, *Solvable Models in Quantum Mechanics*, Theoretical and Mathematical Physics, Springer, 1988, [doi:10.1007/978-3-642-88201-2](https://doi.org/10.1007/978-3-642-88201-2).